Perneti Tarun Venkata Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.51
Sri chaitanya collage	Class XII (BIEAP)	2023	96.00%
Sri chaitanya school	Class X (BSEAP)	2021	99.66%

SKILLS

Programming Languages: C, C++, Python.

Web Technologies: HTML, CSS, JavaScript, React, Node.js, Express.js.

Databases: MySQL, MongoDB.

Libraries & Tools: NumPy, Pandas, Git, GitHub, Solid Edge,Abaqus,Ansys Fluent.

RELEVANT COURSES

Computer Science: Data Structures and Algorithms, Database Management Systems-I, Introduction to Programming.

Mathematics: Calculus-I, Calculus-II, Elementary Linear Algebra, Differential Equations, Transform Techniques, Introduction to Probability, Introduction to Statistics, Complex Variables,

PROJECTS

Autonomous E-Commerce Refund Gatekeeper

LangGraph, LangChain, Gemini API, Multi-Agent Architecture, Streamlit

- Architected a stateful **multi-agent system in LangGraph** powered by **Gemini API**, orchestrating specialized agents (Policy, Risk, Resolution) to autonomously process refund claims **for refund automation**.
- Implemented **conditional routing and human-in-the-loop execution**, auto-approving low-risk refunds while safely pausing high-value claims for human oversight **with configurable risk thresholds and escalation**.
- Developed an interactive **dashboard** providing live workflow visibility and continuous state synchronization for seamless manager approvals **with real-time monitoring and approval tracking for managers**.

Explainable Credit Risk Prediction System

Python, scikit-learn, XGBoost/LightGBM/CatBoost, SHAP, Streamlit

- Architecting a leakage-safe **ML pipeline for credit-default prediction** on the Home Credit Default Risk dataset, with config-driven data validation, feature engineering, and a preprocessing pipeline fit strictly on training data to prevent leakage.
- Building and benchmarking **gradient-boosted models** (Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost) on imbalanced data, evaluating with ROC-AUC, PR-AUC, and calibration metrics, achieving a ROC-AUC of 0.74.
- Developing a multi-page **Streamlit dashboard with SHAP explainability** for individual risk predictions.

Dataset Distillation using LoRA and PEFT Methods (MLPS Course Project)

Machine Learning, Deep Learning, Dataset Distillation, Parameter-Efficient Fine-Tuning

- Conducted extensive research on dataset distillation and **PEFT by integrating LoRA with distillation paradigms** on MobileNetV2 architectures.
- Implemented and evaluated **dataset distillation frameworks including meta-learning, gradient matching, trajectory matching (MTT), and SRe2L statistical recovery** on CIFAR-10 with custom LoRA-based convolutional layers.
- Developed a comprehensive **experimental pipeline involving LoRA integration, gradient tracking ablations,**

synthetic dataset optimization, and cross-evaluation analysis to study compatibility of low-rank adaptation with dataset distillation techniques.
- Analyzed **performance degradation and representational instability in low-rank constrained models**, demonstrating limitations of combining LoRA with dataset distillation.

DBInsight - Database Performance & Analytics System | **MySQL, SQL**

Database Design, Normalization, Indexing, Query Optimization, Transactions, ACID

- Designed and implemented a **normalized relational database** with **10+ interconnected tables** using ER modeling, primary/foreign keys, constraints, and referential integrity to manage customers, orders, inventory, payments, and shipments.
- Developed and optimized **50+ complex SQL queries** using **CTEs, window functions, JOINs, subqueries, B-tree/composite indexes, and EXPLAIN**, improving query execution performance by **~40%** on benchmark datasets.
- Implemented **ACID-compliant transactions, concurrency controls, database roles, access privileges, stored procedures, and triggers** to ensure data consistency, security, and reliable database operations.

EXTRACURRICULAR ACTIVITIES

NSS Volunteer, IIT Hyderabad

- Participated in campus and community development programs, including cleanliness drives and awareness campaigns
- Engaged in regular physical fitness and mental training activities, fostering discipline, resilience, and focus

POSITIONS OF RESPONSIBILITY

Undergraduate Mentor - Sunshine | 2024-25

- Organized and managed departmental orientation programs, mentor-mentee meetings, and student engagement events.

Logistics Coordinator - Media Council

- Coordinated event logistics and resource management, ensuring timely execution and smooth on-ground operations.